

# GAURI MAKKER

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## Education

### Vellore Institute of Technology (VIT) Bhopal

B.Tech in Computer Science and Engineering; CGPA: 8.44/10

Bhopal, India

Aug 2023 – May 2027

## Experience

### Neuronafence

Feb 2025 – Jul 2025

AI Engineer Intern

- Built a retrieval-augmented generation (RAG) chatbot over 10,000+ internal company documents using Python, LangChain and Hugging Face Transformers.
- Designed the document ingestion pipeline (parsing, chunking, embedding) so the knowledge base could be updated without retraining.

### Caliche Global

May 2024 – Jul 2024

Software Development & ML Intern

- Built REST APIs in Flask for a farm management system, including an ML model that provides soil-health insights.
- Wrote SDLC documentation and worked in Agile sprints with the client team.

## Projects

### PlayVision: Football Analytics with Computer Vision

[github.com/makkergauri/football\\_player\\_stats\\_tracker](https://github.com/makkergauri/football_player_stats_tracker)

Python, PyTorch, YOLOv8, ByteTrack, SigLIP, OpenCV

- Team project: detected players, referees and the ball with YOLOv8 and tracked identities across frames with ByteTrack.
- Identified teams without manual labels by clustering SigLIP embeddings of player crops (UMAP); mapped positions to a top-down pitch via a keypoint model and perspective transform.
- Computed possession, player speed and territory control, displayed in a tactical radar view.

### LawBot: RAG-based Legal Assistant

[github.com/makkergauri/lawbot](https://github.com/makkergauri/lawbot)

Python, LangChain, FAISS, Ollama, Groq, Streamlit

- Team project: built a question-answering assistant that answers only from uploaded legal PDFs, with prompts designed to refuse answers not grounded in the documents.
- Implemented PDF extraction, overlapping recursive chunking, nomic-embed-text embeddings and a persistent FAISS index; supports a local LLM (Ollama) or a cloud LLM (Groq API).

### CommBot: Multilingual Offline Disaster Alerts

[github.com/makkergauri/commbot](https://github.com/makkergauri/commbot)

Python, LLMs, Twilio, IVR

- Ingests official CAP warnings (NDMA SACHET, IMD) and delivers them by SMS and phone call in each recipient's language, so people without internet access still receive warnings.
- Restricted the LLM to choosing from fixed labels and copying text from the source alert; every extracted fact is validated against the original, and messages are built from reviewed templates.

### Path Tracer from Scratch

[github.com/makkergauri/raytracer](https://github.com/makkergauri/raytracer)

C++, OpenMP, CMake

- Physically based renderer with no graphics libraries: diffuse, metal and glass materials, BVH acceleration, depth of field and ACES tone mapping, multi-threaded with OpenMP.

## Technical Skills

**Languages:** Python, Java, C++, JavaScript, TypeScript, SQL

**ML & AI:** PyTorch, TensorFlow, OpenCV, YOLOv8, MediaPipe, Hugging Face Transformers, LangChain, FAISS, Ollama, NumPy, Pandas

**Backend & Cloud:** Flask, REST APIs, Streamlit, MySQL, AWS

**Tools:** Git, GitHub, Linux, CMake, Jupyter Notebook, VS Code

**Certifications:** AWS Certified Cloud Practitioner (CLF-C02); NPTEL Introduction to Machine Learning (IIT Madras); AWS Technical Essentials; HackerRank Problem Solving (Intermediate)

## Leadership & Awards

### Notion Community Club, VIT Bhopal

Jan 2024 – Mar 2026

President (Jul 2025 – Mar 2026); earlier Operations Manager and Creative Team Lead

Bhopal

- Led a 130+ member club; organized 10+ workshops and 2 hackathons with 300+ total participants.

### Institution's Innovation Council (IIC), VIT Bhopal

Apr 2025 – Sep 2026

Student Internship Coordinator

Bhopal

- Coordinated student participation in Ministry of Education innovation programs: startup ideation contests, IPR sessions and industry mentorship.

### INSPIRE-MANAK National Level Exhibition and Project Competition (NLEPC)

2019

Government of India; advanced through the district and state rounds to the national exhibition